

2026.4.14. ODE II.

0. 上次讲义的积分因子部分.

$$y' + p(x)y = 0 \Rightarrow y = C e^{-\int_a^x p(t) dt}$$

1. 一阶线性 ODE. 常数变易法. 非常重要!

$$\textcircled{1} y' + p(x)y = q(x).$$

ANSATZ: $y = C(x) e^{-\int_a^x p(t) dt}$

猜测解的形式
在一般方程求解

$$y' = e^{-\int_a^x p(t) dt} (C'(x) - C(x)p(x)).$$

中很重要.

$$= q(x) - p(x)y.$$

对比已知 $C'(x) = q(x) e^{\int_a^x p(t) dt}$

积分: $C(x) = \int_a^x q(t) e^{\int_a^t p(s) ds} dt + C.$

Eg. $y' + y = x^2.$

Step 1. 求解齐次方程 $y' + y = 0.$

$$y = C e^{-x}$$

Step 2. 常数变易.

$$y = C(x) e^{-x}$$

$$y' = C'(x) e^{-x} - y = x^2 - y$$

$$\Rightarrow C'(x) = x^2 e^x.$$

$$C(x) = e^x (x^2 - 2x + 2) + C$$

$$\text{因此, } y(x) = x^2 - 2x + 2 + C e^{-x}.$$

Rmk. 另一种理解: $y = y_1 + y_2$

其中, y_1 满足 $y_1' + y_1 = x^2$. y_2 是 $y' + y = 0$ 通解.

若仅猜测 y_1 , 容易发现应是个二次函数.

$$\text{令 } y_1 = x^2 + ax + b. \quad y_1' = 2x + a.$$

$$y_1 + y_1' = x^2 + (a+2)x + (a+b) = x^2$$

$$\Rightarrow \begin{cases} a+2=0 \\ a+b=0 \end{cases} \Rightarrow \begin{cases} a=-2 \\ b=2 \end{cases} \quad y_1 = x^2 - 2x + 2.$$

Eg. $xy' - 2y = 2x^4$.

Step 1. $y' - \frac{2}{x}y = 0$ 齐次方程 $\Rightarrow y = Cx^2$.

Step 2. $y = C(x)x^2$. $y' = C'(x)x^2 + 2xC(x)$

$$xy' - 2y = C'(x)x^3 + 2x^2C(x) - 2x^2C(x)$$

$$= C'(x)x^3 = 2x^4 \Rightarrow C'(x) = 2x, \quad C(x) = x^2 + C.$$

$$y = x^4 + Cx^2.$$

另一种理解: $y = Cx^2 + y_0(x)$. 满足 $xy_0' - 2y_0 = 2x^4$.

应猜 $y_0(x) = x^4 + ax^3 + bx^2 + cx + d$.

$$y_0'(x) = 4x^3 + 3ax^2 + 2bx + c.$$

$$xy_0' - 2y_0 = 2x^4 + (3a-2a)x^3 + (2b-2b)x^2 + (c-2c)x +$$

$$\Rightarrow \begin{cases} a=0 \\ b \text{ 任意} \\ c=0 \\ d=0 \end{cases} \Rightarrow y_0(x) = x^4 + \widetilde{C}x^2.$$

$$\Rightarrow y(x) = x^4 + Cx^2.$$

② 变式: Bernoulli.

$$y' + p(x)y = q(x)y^\alpha. \quad (y \neq 0, 1).$$

$$\text{令 } z = y^{1-\alpha}. \quad z' = (1-\alpha)y^{-\alpha}y'.$$

$$\Rightarrow y^{-\alpha}y' + p(x)y^{1-\alpha} = q(x).$$

$$\Rightarrow z' + (1-\alpha)p(x)z = (1-\alpha)q(x). \text{ 变成一次的.}$$

Eg. (2023.1.) $(xy - x^3y^3)dx + (1+x^2)dy = 0. \quad y(0)=1$

$$\Leftrightarrow y' = \frac{x^3y^3 - xy}{1+x^2} \Leftrightarrow y' + \frac{x}{1+x^2}y = \frac{x^3}{1+x^2}y^3.$$

Rmk. 很容易想到换元 $t = xy$. 但不好做!

谨记! 人们一般只会解关于 y 线性的 ODE.

如果不能分离变量, 且 y 和 y^α 同时出现, 优先考虑 Bernoulli ODE.

$$\text{令 } z = y^{1-3} = y^{-2}. \quad z' = -2y^{-3}y'.$$

$$\text{原式: } -2y^{-3}y' - \frac{2x}{1+x^2}y^{-2} = \frac{-2x^3}{1+x^2}$$

$$\text{即 } \underbrace{z' - \frac{2x}{1+x^2}z}_{\text{积分因子}} = -\frac{2x^3}{1+x^2}.$$

$$\text{积分因子 } \rightarrow \int \frac{2x dx}{1+x^2} = \ln(1+x^2) \quad e^{\int \frac{2x dx}{1+x^2}} = \frac{1}{1+x^2}$$

$$\frac{1}{1+x^2}z' - \frac{2x}{(1+x^2)^2}z = \left(\frac{z}{1+x^2}\right)'$$

$$\Rightarrow \left(\frac{z}{1+x^2}\right)' = -\frac{2x^3}{(1+x^2)^2}$$

$$\begin{aligned} \Rightarrow \frac{z}{1+x^2} + C &= -\int \frac{2x^3 dx}{(1+x^2)^2} = \int \frac{-x^2 dx^2}{(1+x^2)^2} = -\int \frac{dx^2}{1+x^2} + \int \frac{dx^2}{(1+x^2)^2} \\ &= -\ln(1+x^2) - \frac{1}{1+x^2} \end{aligned}$$

$$\Rightarrow y^{-2} = z = -1 - (1+x^2)\ln(1+x^2) + C(1+x^2).$$

代入 $\lambda(x, y) = (0, 1)$ 知 $1 = -1 - 0 + C$ 系 $C = 2$

$\Rightarrow (2x^2 + 1 - (1+x^2)\ln(1+x^2)) y^2 = 1.$

2. 降阶. 即关于 y 的高阶导数.

① 仅含 $y^{(k)}, y^{(k+1)}, x$.

换元 $p = y^{(k)}$. 变成 p', p, x 的方程.

再逐步积分回 y .

Eg. 本次作业. $y''' = 2(y'' - 1) \cot x$.

令 $p = y''$. $p' = 2(p - 1) \cot x$. 即求出 y''

积分 2 次即得 y .

Eg. $x^2 y'' = (y')^2$ 令 $p = y'$.

$x^2 p' = p^2$. 即 $p' = \left(\frac{p}{x}\right)^2$ 是 Bernoulli 方程

~~看出 $p(x) = ax^\alpha$. $p' = \alpha ax^{\alpha-1} = \left(\frac{p}{x}\right)^2 = a^2 x^{2\alpha-2}$~~

~~$\Rightarrow \alpha = a = 1$~~

~~即 $p = x$. 是特解.~~

~~那么, $y' = x \Rightarrow$~~

~~又 $p' = 0$ 通解为 $p = C$. 故 $p = x + C$~~

令 $z = p'^{-2} = p^{-1}$. $\Rightarrow z' = -\frac{1}{x^2}$

$\Rightarrow z = \frac{1}{x} + C \Rightarrow p = \frac{1}{\frac{1}{x} + C} = \frac{x}{1 + Cx}$

那么, $y' = \frac{x}{1 + Cx}$, $dy = \frac{x}{1 + Cx} dx$

$= \frac{1}{C} \left(\frac{Cx+1}{1+Cx} - \frac{1}{1+Cx} \right) dx$

$= \frac{1}{C} - \frac{1}{C} \cdot \frac{dx}{1+Cx}$

$\Rightarrow y = \frac{x}{C} - \frac{1}{C^2} \ln(1+Cx) + D$.

② 仅含 y, y', y'' . (自治系统).

$$\text{换元 } y' = p. \quad y'' = \frac{dp}{dx} = \frac{dp}{dy} \cdot y' = p \frac{dp}{dy}.$$

变成关于 y, p 的方程. (以 y 为变元!).

$$\text{Eg } y'^2 + 2yy'' = 0.$$

$$\text{令 } p = y'. \quad p^2 + 2y \cdot p \frac{dp}{dy} = 0.$$

$$(i) p \equiv 0 \sim y = C.$$

$$(ii) p + 2y \frac{dp}{dy} = 0.$$

$$\Rightarrow \sqrt{y} \cdot \frac{dp}{dy} + \frac{1}{2\sqrt{y}} p = 0. \Rightarrow \frac{d}{dy}(\sqrt{y} p) = 0.$$

$$\Rightarrow p = C y^{-\frac{1}{2}}. \quad \text{即 } y' = C y^{-\frac{1}{2}}$$

$$\text{即 } y' + C y^{-\frac{1}{2}} = 0. \quad \text{这是 Bernoulli 方程.}$$

$$\text{令 } z = y^{1 - (-\frac{1}{2})} = y^{\frac{3}{2}}. \quad z' = \frac{3}{2} y^{\frac{1}{2}} y'.$$

$$\Rightarrow z' = \frac{3}{2} y^{\frac{1}{2}} y' = \frac{3}{2} C = C.$$

$$\Rightarrow z = Cx + D \quad \text{即 } y = (Cx + D)^{\frac{2}{3}}.$$

($y = C$ 也包含其中).

$$\text{综上: } y = (Cx + D)^{\frac{2}{3}}.$$

3. 积分因子方法. 以下均写为 $Pdx + Qdy = 0$ 形式.

$$\text{① } (x^2 + y^2 + x) dx + y dy = 0. \quad \text{需要对 } \frac{\partial}{\partial x} \text{ 非 } 0. \text{ 考虑 } f(x) \text{ 型积分因子.}$$

$$\frac{\partial P}{\partial y} = 2y, \quad \frac{\partial Q}{\partial x} = 0.$$

$$\frac{\partial}{\partial y}(P f(x)) = 2y f(x)$$

$$\frac{\partial}{\partial x}(Q f(x)) = y f'(x)$$

$$\text{令之相等. } \Leftrightarrow 2f = f' \quad \Leftrightarrow f(x) = e^{2x}.$$

$$\Rightarrow e^{2x}(x^2+y^2+x)dx + e^{2x}y dy = 0.$$

$$\Rightarrow \frac{1}{2}e^{2x}y^2 + \frac{1}{2}e^{2x}x^2 = C. \text{ 即 } e^{2x}(x^2+y^2) = C.$$

$$\textcircled{2} (x^2+y^2+y)dx - xdy = 0.$$

$$\frac{\partial P}{\partial y} = 2y+1, \quad \frac{\partial Q}{\partial x} = -1.$$

直接用 $f(x), f(y)$ 都不行.

有 x^2+y^2 . 考虑 $f(x^2+y^2)$.

$$\text{除 } x^2+y^2 \Rightarrow \left(1 + \frac{y}{x^2+y^2}\right)dx - \frac{x}{x^2+y^2}dy = 0.$$

$$\frac{\partial P}{\partial y} = \frac{(x^2+y^2)-2y^2}{(x^2+y^2)^2} = \frac{x^2-y^2}{(x^2+y^2)^2} = \frac{\partial Q}{\partial x}$$

$$\Rightarrow x + \arctan \frac{x}{y} = C.$$

一般 $xy, \frac{1}{xy}, x^2+y^2, \frac{1}{x^2+y^2}$

Rmk. 常见积分因子: $f(x), f(y), f(xy), f(x^2+y^2), x^a y^b$.

$$\textcircled{3} y dy = (x dy + y dx) \sqrt{1+y^2} \rightarrow \text{除掉 } \sqrt{1+y^2}.$$

$$\left(x - \frac{y}{\sqrt{1+y^2}}\right)dy + y dx = 0. \quad \frac{\partial P}{\partial x} = 1 = \frac{\partial Q}{\partial y}.$$

$$\Rightarrow \frac{1}{2}(x^2+y^2) - \sqrt{1+y^2} = C.$$

$$\textcircled{4} y^2 dx - (xy + x^3) dy = 0.$$

总是先试 $f(x), f(y)$.

$$\frac{\partial P}{\partial y} = 2y, \quad \frac{\partial Q}{\partial x} = -y - 3x^2.$$

$$\frac{\partial}{\partial y}(Pf(x)) = 2yf(x). \quad \frac{\partial}{\partial x}(Qf(x)) = -xyf'(x) - yf(x) - x^3f'(x) - 3x^2f(x).$$

$$\text{令之相等} \Leftrightarrow 3(y+x^2)f(x) = -x(y+x^2)f'(x) \quad \underline{y = -x^2 \text{ 不是解}}$$

$$\Leftrightarrow f(x) = -\frac{x}{3}f'(x) \Leftrightarrow f' + \frac{3}{x}f = 0$$

$$\Leftrightarrow (x^3f)' = 0. \Leftrightarrow f(x) = \frac{1}{x^3} \text{ 即可.}$$

$$\Rightarrow \frac{y^2}{x^3} dx - \left(1 + \frac{y}{x^2}\right) dy = 0 \Rightarrow \frac{y^2}{x^2} + 2y = C.$$

$$\textcircled{5} \quad y dx - x dy = 2x^3 \tan \frac{y}{x} dx.$$

无法处理, 令 $z = \frac{y}{x}$.

$$dz = \frac{x dy - y dx}{x^2}$$

$$\Rightarrow -dz = 2x \tan z dx \quad \text{分离变量.}$$

$$dx^2 = -\cot z dz = -d \ln |\sin z|.$$

$$\Rightarrow -x^2 + C = \ln |\sin z| \Rightarrow \sin \frac{y}{x} = Ce^{-x^2}, \quad C \in \mathbb{R}.$$

$$\textcircled{6} \quad xy dx - (y^3 + x^2y + x^2) dy = 0. \quad \text{只需要一个 } -xy + 1$$

$$\frac{\partial P}{\partial y} = x, \quad \frac{\partial Q}{\partial x} = -2x(y+1). \quad \text{找 } f(y)!$$

$$\frac{\partial}{\partial y}(Pf(y)) = x(f + yf'). \quad \frac{\partial}{\partial x}(Qf(y)) = -2x(y+1)f$$

$$\text{令相等} \Leftrightarrow f + yf' = -2(y+1)f$$

$$\Leftrightarrow f' + (2 + \frac{3}{y})f = 0. \quad \int 2 + \frac{3}{y} dy = y^3 e^{2y}$$

$$\Leftrightarrow (y^3 e^{2y} f)' = 0 \Rightarrow f(y) = y^{-3} e^{-2y} \checkmark.$$

$$\Rightarrow xy^{-2} e^{-2y} dx - y^{-3} e^{-2y} (y^3 + x^2y + x^2) dy = 0$$

$$\Rightarrow \frac{1}{2} x^2 y^{-2} e^{-2y} - \frac{1}{2} e^{-2y} = C \Rightarrow \frac{x^2 - y^2}{y^2} = Ce^{2y}.$$

$$\textcircled{7} \quad (x^2 - y^2 + y) dx + x(2y - 1) dy = 0.$$

$$\frac{\partial P}{\partial y} = 1 - 2y, \quad \frac{\partial Q}{\partial x} = 2y - 1. \quad \text{只需要 } -1. \text{ 与 } y \text{ 无关. 找 } f(x).$$

$$\frac{\partial}{\partial y}(Pf(x)) = (1 - 2y)f(x). \quad \frac{\partial}{\partial x}(Qf(x)) = (2y - 1)(xf' + f).$$

$$\text{令 } f + xf' + f = 0. \quad \text{即 } f' + \frac{2}{x}f = 0. \quad e^{\int \frac{2}{x} dx} = x^2.$$

$$\Rightarrow (x^2 f)' = 0 \Rightarrow f(x) = \frac{1}{x^2} \checkmark.$$

$$\Rightarrow (1 - \frac{y^2}{x^2} + \frac{y}{x^2}) dx + \frac{2y-1}{x} dy = 0 \Rightarrow \frac{y^2-y}{x} + x = C \quad \text{即 } x^2 + y^2 - y = Cx.$$